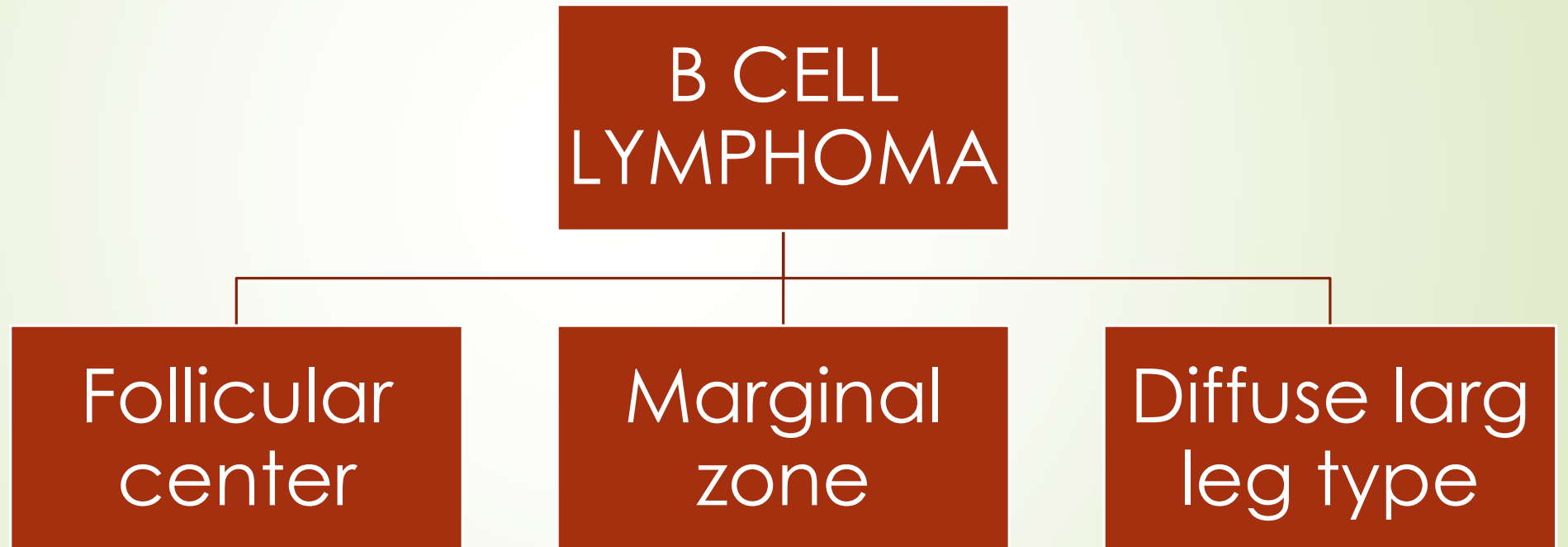


B CELL LYMPHOMA BCC

1

Dr eman gomaa
Consutant dermatologist

B CELL LYMPHOMA



CLASSIFICATION OF B-CELL LYMPHOMAS WITH PRIMARY CUTANEOUS MANIFESTATIONS

WHO-EORTC 2005	WHO 2008/2016
• Primary cutaneous follicle center lymphoma • Primary cutaneous marginal zone B-cell lymphoma* • Primary cutaneous diffuse large B-cell lymphoma, leg type • Primary cutaneous diffuse large B-cell lymphoma, other • Intravascular diffuse large B-cell lymphoma	• Primary cutaneous follicle center lymphoma • Extranodal marginal zone lymphoma of <i>mucosa-associated lymphoid tissue</i> – MALT lymphoma • Primary cutaneous diffuse large B-cell lymphoma, leg type • Diffuse large B-cell lymphoma, NOS • Intravascular diffuse large B-cell lymphoma



Fig. 119.1 Cutaneous follicle center lymphoma. Prominent pink-violet nodule on the lateral forehead with a few telangiectasias. The nodule is surrounded by papules and large infiltrated plaques.



Fig. 119.2 Cutaneous follicle center lymphoma. Large nodule on the upper back. Note the surrounding erythematous papules, patches, and plaques (Crosti's lymphoma).



Fig. 119.3 Cutaneous marginal zone B-cell lymphoma. Two well-circumscribed, erythematous nodules on the shoulder.



Fig. 119.4 Cutaneous marginal zone B-cell lymphoma. Two thick pink-violet plaques on the foot, one of which has a flat surface. Molecular analysis revealed *Borrelia* DNA within the infiltrate. In the past, these tumors, which demonstrated prominent lymphoplasmacytic differentiation histologically, were classified as cutaneous immunocytomas.



Fig. 119.5 Cutaneous diffuse large B-cell lymphoma, leg type. Multiple coalescing red-brown papulonodules and plaques on the lower leg.

	Follicular center	Marginal zone	Diffuse larg leg type
	Neoplastic proliferation of mature germinal center B cell	Neoplastic proliferation of marginal zone b cell	Lymphoma composed of larg rounded b cell
	Centrocyte centroblast	Centrocyte like or monocytoid	Centroblast , immunoblast
	Common 60% men	25% 50yrs old men	10-20% elderly women 70yrs old
symptom	Fever , night sweat , weight loss		

	Follicular center	Marginal zone	Diffuse larg leg type
	Scalp , forehead ,	Up-extrematis , trunk	Distal spect of one leg
	Pink to plum papule , plaque ,tumor with peripheral halo of erythema (crostis lymphoma) Rare ulceration	Pink to violet papule , plaque , nodule >>normal surrounding skin 2ry anetoderma Rare ulceration	Erythamatus red , brown nodules >>> ulceration
H.P	Nodular or diffuse infeltirate all dermis , s.c Spared epidermis Lymphoid follicle with atypical feature	Nodular or diffuse infiltrat all dermis , sc Spared epidermis	
	SCE Dermatology / EBDV by dr :Eman - CoNnect Academy CD20+ , CD79 + , PAX5 +		

	Follicular center	Marginal zone	Diffuse larg leg type
	BCL6 +ve , CD10 +ve	BCL2 +ve , MuM +ve	BCL2 +ve , MUM +VE
	Centrocyte	Dutcher bodies	Immunoblast , centroblast
prognosis	Favorable	Excellent	bad
DD			

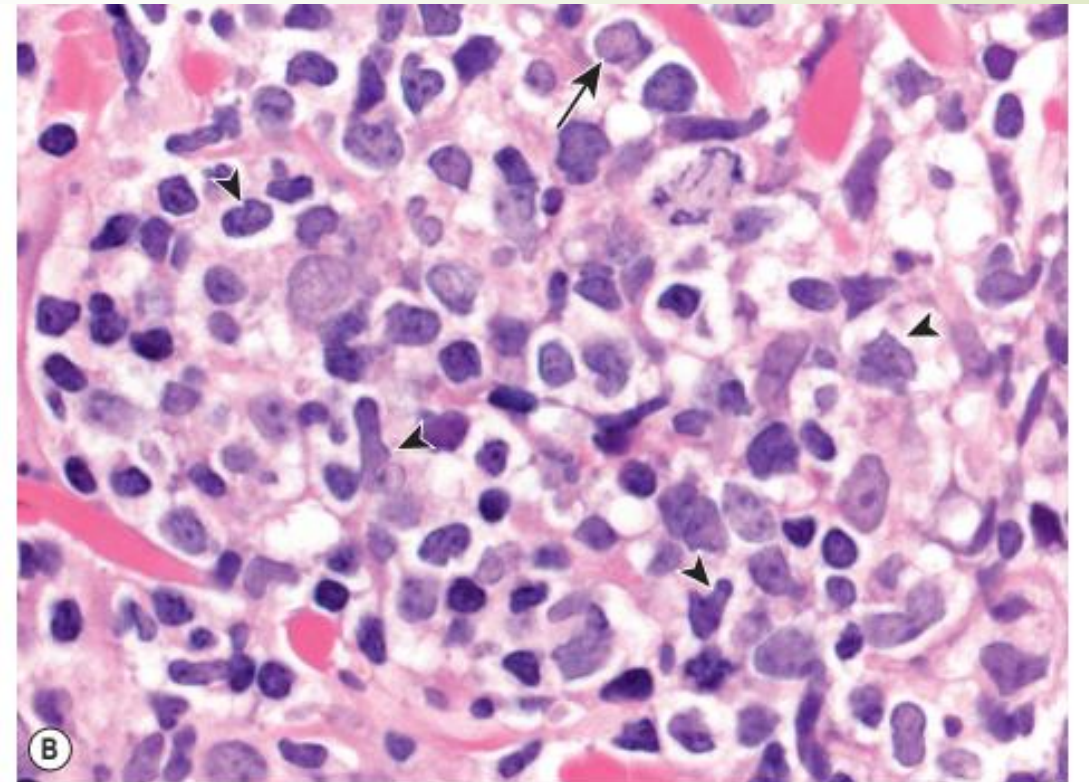
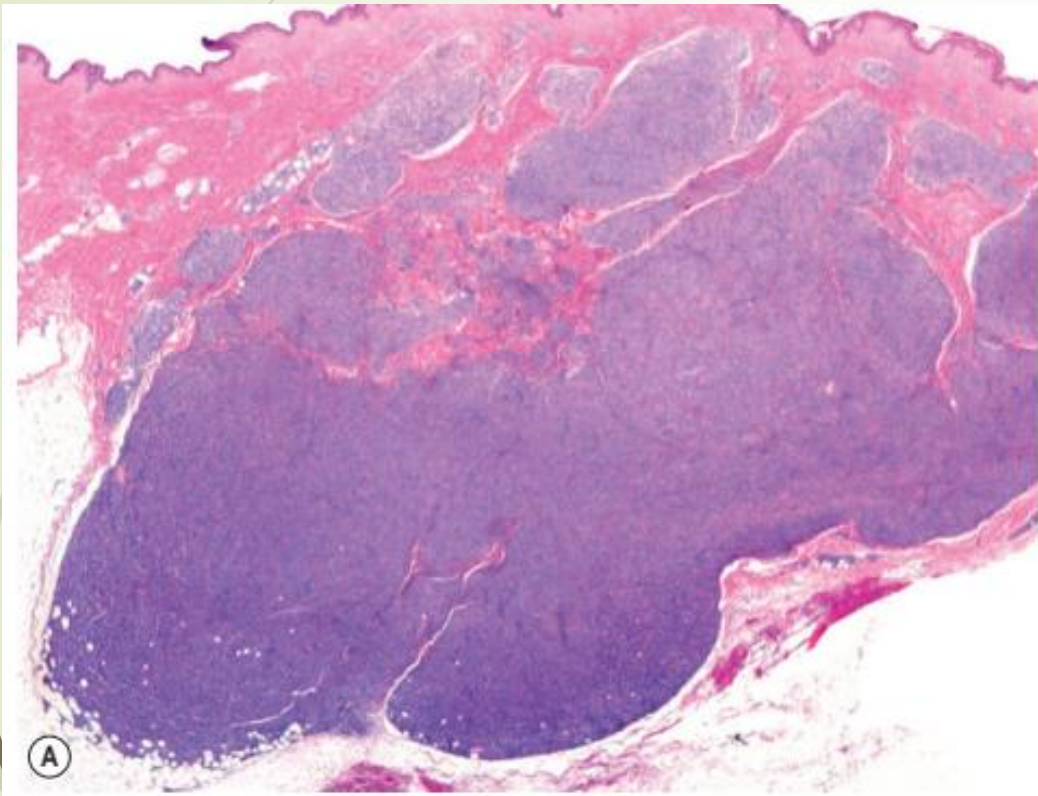


Fig. 119.6 Cutaneous follicle center lymphoma, diffuse type. **A** Diffuse infiltrate without a follicular pattern. **B** Centroblasts (arrow) and medium- and large-sized centrocytes (cleaved cells; arrowheads) predominate.

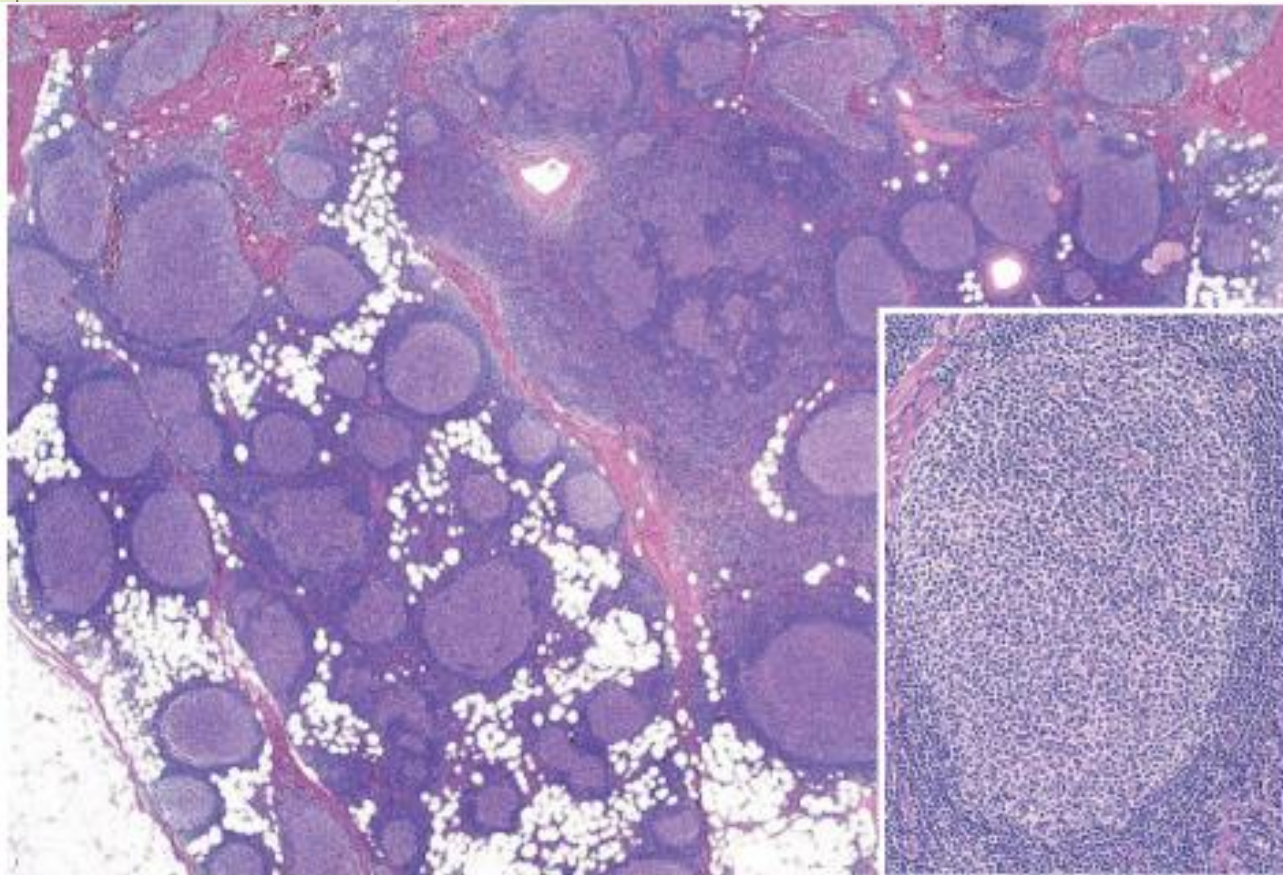


Fig. 119.7 Cutaneous follicle center lymphoma, follicular type. Neoplastic follicles with a monomorphous morphology (lack of polarization). Note the absence of dark and clear areas within the central portion of the follicle (inset).

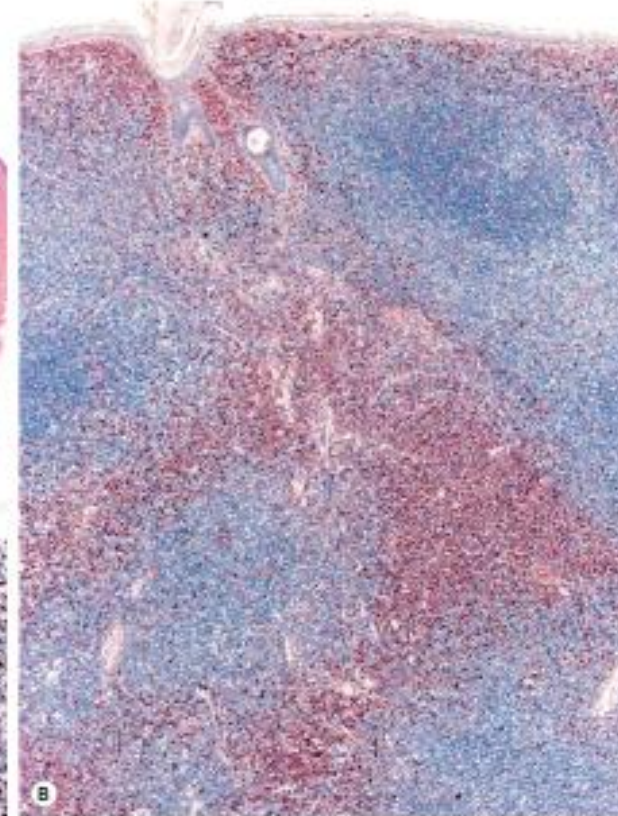
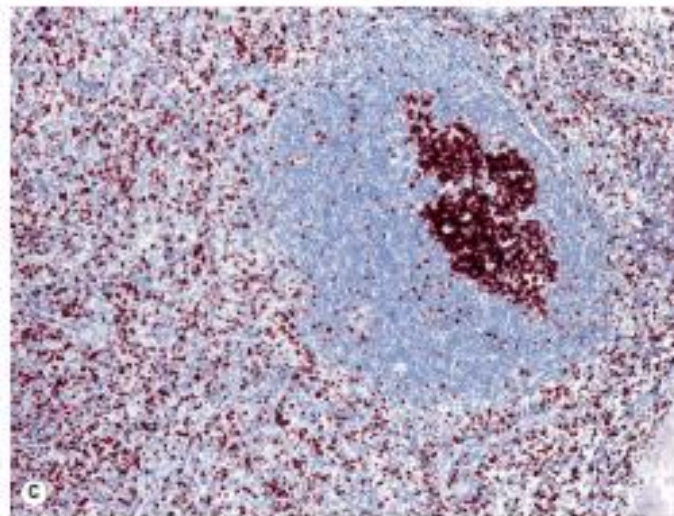
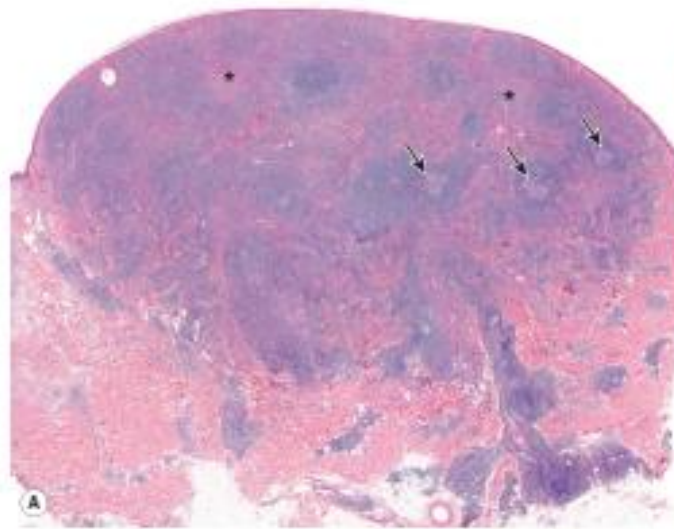


Fig. 119.8 Cutaneous marginal zone B-cell lymphoma. A Small nodules of reactive lymphocytes (dark areas), some of which have reactive germinal centers (arrows). The nodules are surrounded by neoplastic marginal zone cells, lymphoplasmacytoid cells, and plasma cells (clear areas [*]). B Monotypic expression of λ immunoglobulin light chain within the neoplastic population of cells (in the clear areas). C MIB-1 staining shows marked proliferation within a germinal center in a reactive lymphoid follicle and increased proliferation of neoplastic cells in the perifollicular area.

IMMUNOHISTOCHEMICAL MARKER EXPRESSION IN PRIMARY AND SECONDARY CUTANEOUS LARGE B-CELL LYMPHOMAS WITH OVERLAP IN ROUTINE HISTOPATHOLOGIC FINDINGS								
	Immunohistochemical marker expression							
	Bcl-2	Bcl-6	CD10	CD5	MUM-1	Cyclin D1	TdT	EBER-1
<i>Primary cutaneous lymphomas</i>								
Follicle center lymphoma, diffuse	-	+	+/+	-	-	-	-	-
Marginal zone lymphoma, blastoid	+	-	-	-	-	-	-	-
Diffuse large B-cell lymphoma, leg type	+	+/+	-	-	+	-	-	-
<i>Systemic lymphomas with secondary skin involvement</i>								
EBV+ DLBCL	+/+	+/+	+/+	-	+	-	-	+
Nodal DLBCL	+	+/+	+/+	+/+	+/+	-	-	-
Mantle cell lymphoma, blastic	+	+/+	-	+	+/+	+	-	-
CLL/SLL → Richter syndrome	+/+	-	-	+	-	-	-	-
Burkitt lymphoma	-	+	+	-	-	-	-	+
B lymphoblastic lymphoma	+	-	+	-	-	-	+	-
+/+, majority of cases positive; +/+, majority of cases negative.								

RECOMMENDED STAGING INVESTIGATIONS FOR PATIENTS WITH A CONFIRMED DIAGNOSIS OF B-CELL LYMPHOMA INVOLVING THE SKIN

History and physical examination

- Presence/absence of “B” symptoms, including fever, night sweats, weight loss, malaise
- Complete lymph node examination
- Palpation of abdomen for hepatosplenomegaly
- Examination of oral cavity
- History of organ transplantation and/or other immune suppression (e.g. HIV, iatrogenic, congenital)

Laboratory studies

- Complete blood count with differential and platelet count
- Comprehensive serum chemistries including LDH
- Flow cytometry of peripheral blood mononuclear cells*

Imaging studies

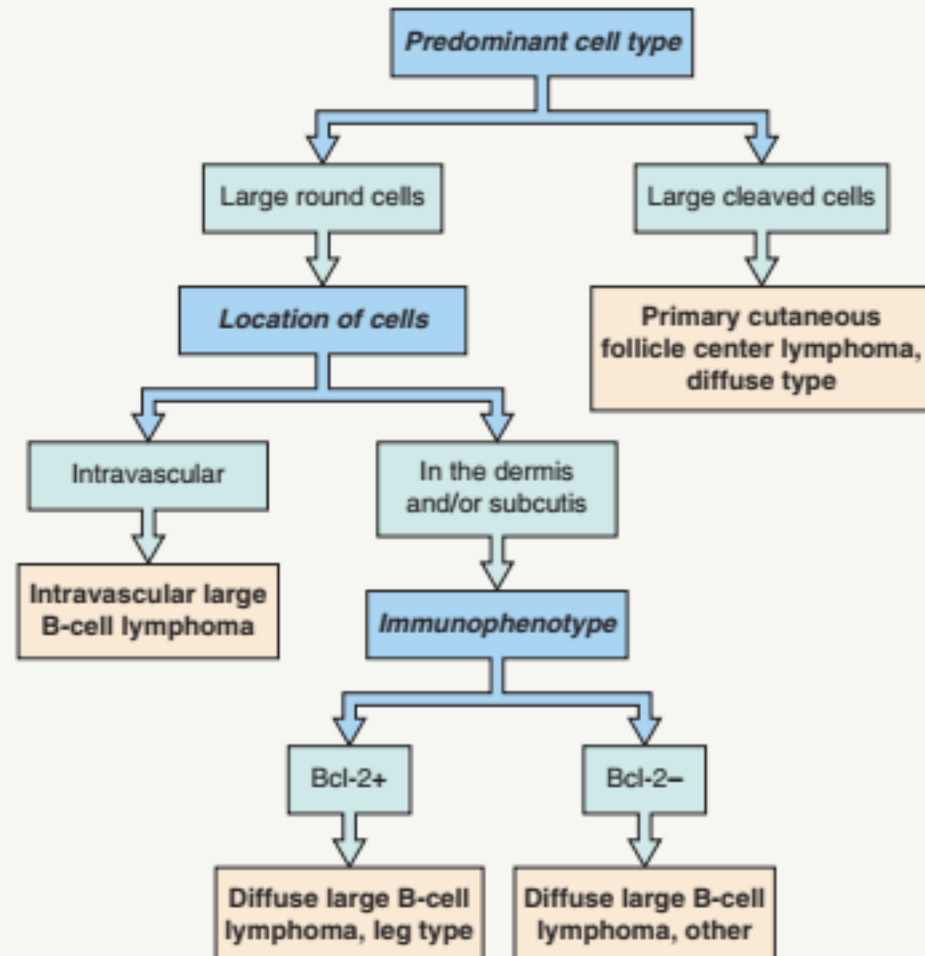
- Ultrasound of abdomen and superficial lymph nodes
- Whole-body PET/CT scan or whole-body CT scan*

Additional studies as clinically indicated

- Bone marrow biopsy**
- Excisional biopsy of enlarged lymph nodes or other suspicious lesions
- PCR analysis for *Borrelia* spp. DNA in patients from endemic areas (Europe) or travelers to endemic areas

	Follicular center	Marginal zone	Diffuse larg
	Low grad indolent BCL • watchfull – waiting strategy • Follow up examination every 6 month • Intralesional triamcinolone • S.c , intralesional interferon alpha • Systemic Ab for borrelia • Local radiotherapy • Surgical • Surgical followed by radiotherapy • RITUXEMAB		Aggressive BCL RITUXIMAB + CHOP
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APPROACH TO THE DIAGNOSIS OF CUTANEOUS LARGE B-CELL LYMPHOMAS



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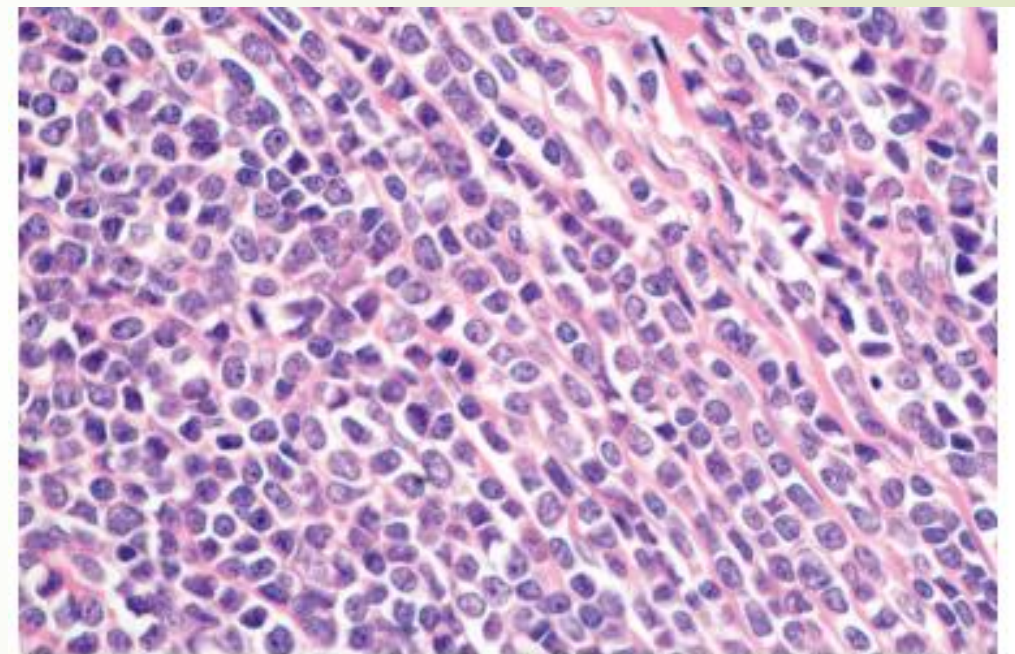
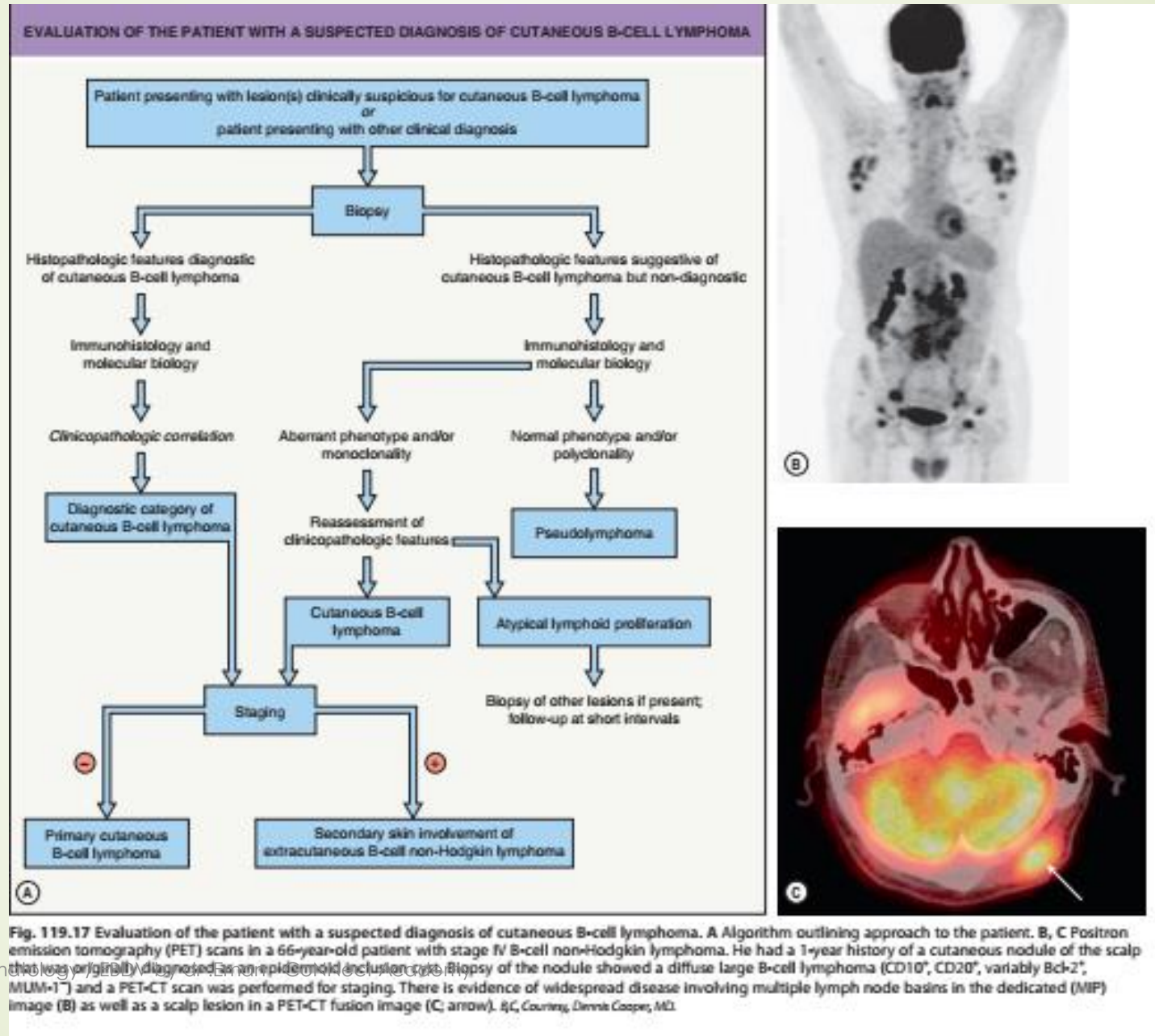


Fig. 119.15 Precursor B lymphoblastic lymphoma/leukemia. Medium-sized blasts with the characteristic "mosaic-stone" linear arrangement.



Online only content



eFig. 119.1 Cutaneous diffuse large B-cell lymphoma, leg type. Multiple red-brown tumors on the lower leg.



eFig. 119.2 Cutaneous follicle center lymphoma. Large ulcerated tumors on the scalp surrounded by infiltrated erythematous nodules and plaques.



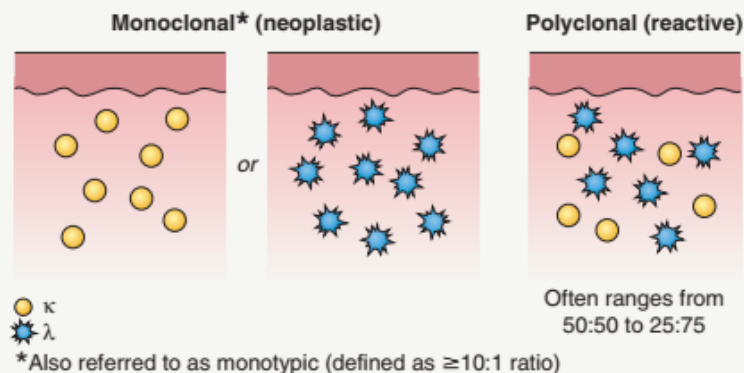
eFig. 119.4 Cutaneous marginal zone B-cell lymphoma. Dome-shaped erythematous nodule with smooth surface. The surrounding area shows features of acrodermatitis chronica atrophicans. This tumor, which demonstrated prominent lymphoplasmacytic differentiation histologically, was classified as cutaneous immunocytoma in the past.



eFig. 119.3 Cutaneous follicle center lymphoma. Large ulcerated tumor on the back. Note surrounding erythematous papules, patches and plaques (Crest's lymphoma).

eFig. 119.5 Cutaneous diffuse large B-cell lymphoma, leg type. Multiple red-brown papulonodules and tumor on the lower leg.

PATTERN OF IMMUNOGLOBULIN LIGHT CHAIN EXPRESSION



CUTANEOUS CONDITIONS ASSOCIATED WITH MONOCLONAL GAMMOPATHY

Proliferation of lymphoplasmacytic cells in the skin

- Extramedullary cutaneous plasmacytoma*
- Cutaneous Waldenström macroglobulinemia†

Deposition of the monoclonal protein in the skin, by definition

- Primary systemic amyloidosis (light chains)
- Cryoglobulinemic occlusive vasculopathy (type I cryoglobulins‡)
- Hyperkeratotic spicules (follicular > non-follicular)
- Crystal-storing histiocytosis
- Crystalglobulinemia
- IgM storage papules (cutaneous macroglobulinosis)†
- Subepidermal bullous dermatosis associated with IgM gammopathy†

Deposition of the monoclonal protein in the skin frequently observed

- Plasma cell dyscrasia-associated acquired cutis laxa, acral or generalized (amyloid and/or IgG)
- Plasma cell dyscrasia-associated reactive angioendotheliomatosis (type I cryoglobulins or amyloid)

Almost always associated with monoclonal gammopathy

- Scleromyxedema
- POEMS syndrome§
- AESOP syndrome = adenopathy and extensive skin patch overlying a plasmacytoma; may coexist with POEMS syndrome§
- Schnitzler syndrome
- Necrobiotic xanthogranuloma

Frequently associated with monoclonal gammopathy

- Normolipemic plane xanthoma
- Scleredema (type 2)
- Angioedema secondary to acquired C1 esterase inhibitor deficiency
- Clarkson syndrome (idiopathic systemic capillary leak syndrome)

Significant association with monoclonal gammopathy, primarily IgA (at least 15% of cases)

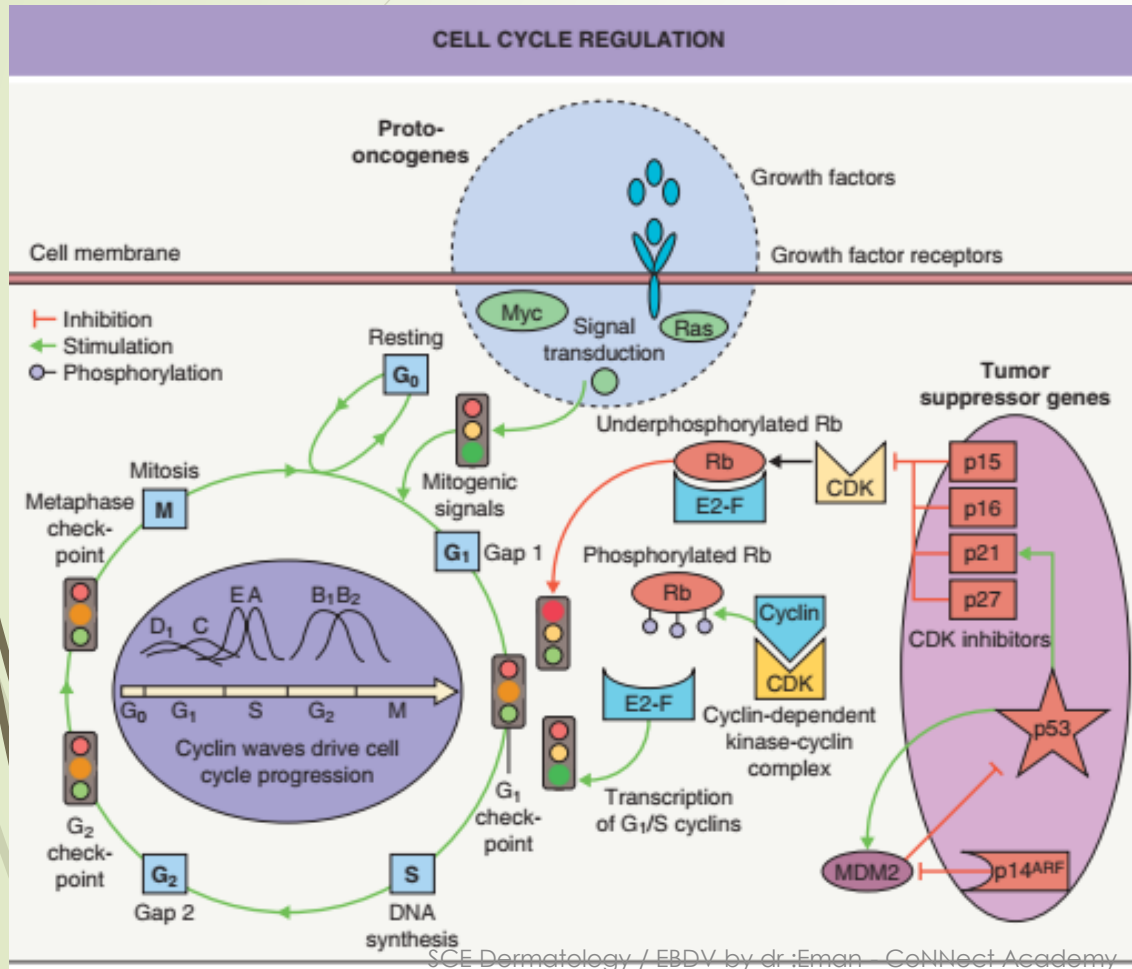
- Erythema elevatum diutinum
- Subcorneal pustular dermatosis (SPD) and SPD-type IgA pemphigus
- Pyoderma gangrenosum

Occasionally associated with monoclonal gammopathy

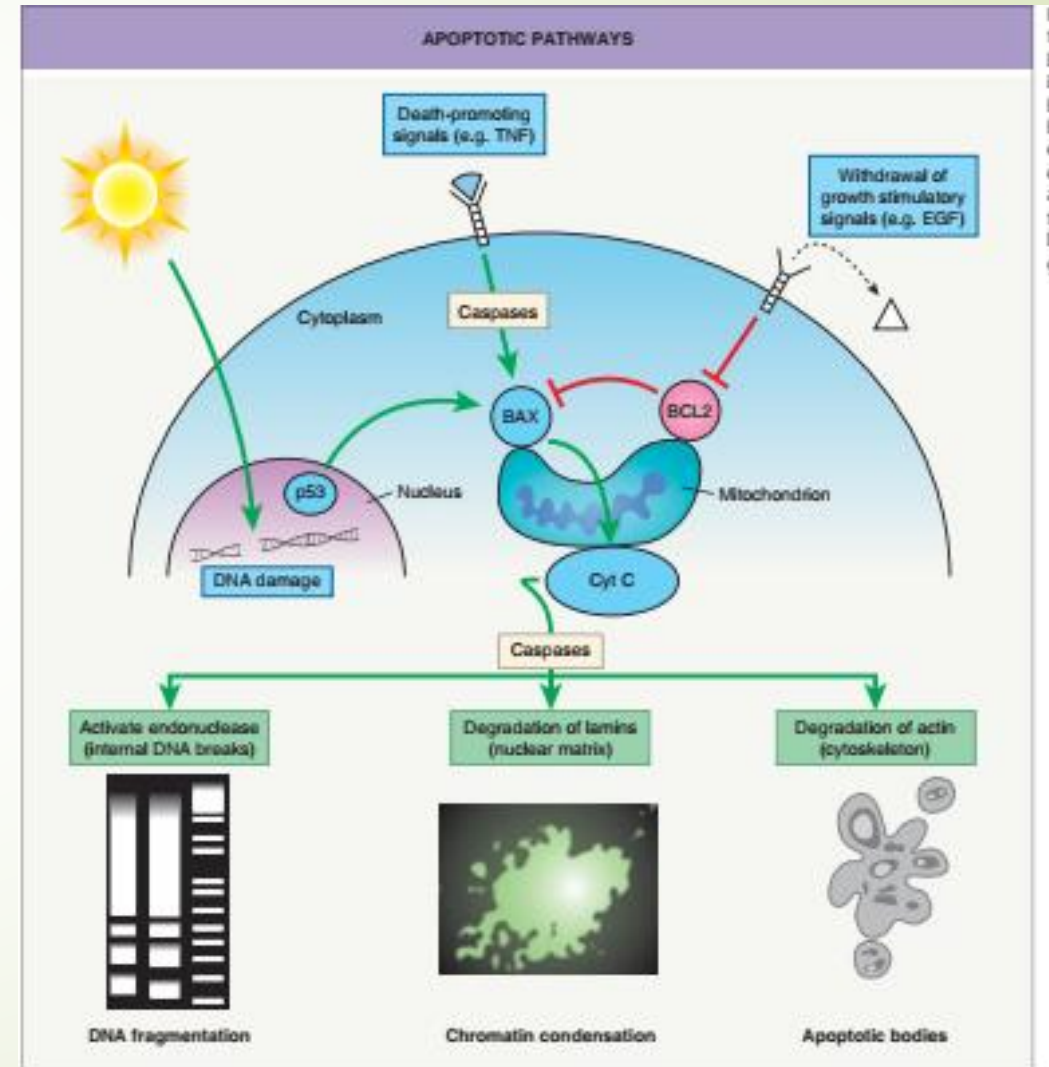
- Sweet syndrome
- Cutaneous small vessel vasculitis‡, including adult IgA vasculitis with monoclonal IgA gammopathy
- Xanthoma disseminatum
- Epidermolysis bullosa acquisita
- Paraneoplastic pemphigus
- Atypical scleroderma¶

Basal cell carcinoma





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EXAMPLES OF ONCOGENES AND TUMOR SUPPRESSOR GENES ASSOCIATED WITH CARCINOGENESIS				
Gene			Protein product	
Name	Function	Chromosome	Location	Function
<i>PDGFB</i>	Oncogene	22	Extracellular	Platelet-derived growth factor (PDGF)
<i>HRAS</i>	Oncogene	11	Membrane	GTPase
<i>SRC</i>	Oncogene	20	Cytoplasm/membrane	Tyrosine kinase
<i>RAF1</i>	Oncogene	3	Cytoplasm/membrane	Serine/threonine kinase
<i>MYC</i>	Oncogene	8	Nucleus	Transcription factor
<i>FOS</i>	Oncogene	14	Nucleus	Transcription factor
<i>RB1</i>	Suppressor gene	13	Nucleus	Cell cycle regulator
<i>TP53</i>	Suppressor gene	17	Nucleus	DNA repair, apoptosis
<i>BCL2</i>	Suppressor gene	18	Mitochondria	Apoptosis

Risk factor for NON melanocytic tumor

RISK FACTORS FOR THE DEVELOPMENT OF BASAL CELL CARCINOMAS (BCCs) AND SQUAMOUS CELL CARCINOMAS (SCCs)		
	SCC	BCC
<i>Environmental exposures</i>		
UV exposure	+	+
Other exposures to UV light (PUVA, tanning beds)	+	+
Ionizing radiation	+	+
Chemicals, including arsenic, mineral oil, coal tar, soot, mechlorethamine (nitrogen mustard), polychlorinated biphenyls, 4,4' bipyridyl, psoralen (plus UVA) ³³	+	+
Human papillomavirus (HPV)	+	(+)
Cigarette smoking	+	
<i>Pigmentary phenotype</i>		
Fair skin, always burns, never tans	+	+
Freckling	+	+
Red hair	+	+
<i>Genetic syndromes</i>		
Xeroderma pigmentosum	+	+
Oculocutaneous albinism	+	+
Epidermodysplasia verruciformis	+	(+)
Dystrophic epidermolysis bullosa (primarily recessive)	+	
Ferguson-Smith syndrome	+	
Muir-Torre syndrome	+*	+*
Basal cell nevus syndrome		+
Bazex-Dupré-Christol and Rombo syndromes		+
<i>Predisposing clinical settings</i>		
Chronic non-healing wounds	+	
Longstanding discoid lupus erythematosus, lichen planus (erosive), or lichen sclerosus	+	
Porokeratosis (especially linear)	+	
Nevus sebaceus	+	+†
<i>Immunosuppression</i>		
Organ transplantation	+	+
Other (e.g. chronic lymphocytic leukemia treated with fludarabine, AIDS patients with HPV infection)	+	(+)

*Both SCCs (keratoacanthoma type) and BCCs typically have sebaceous differentiation.

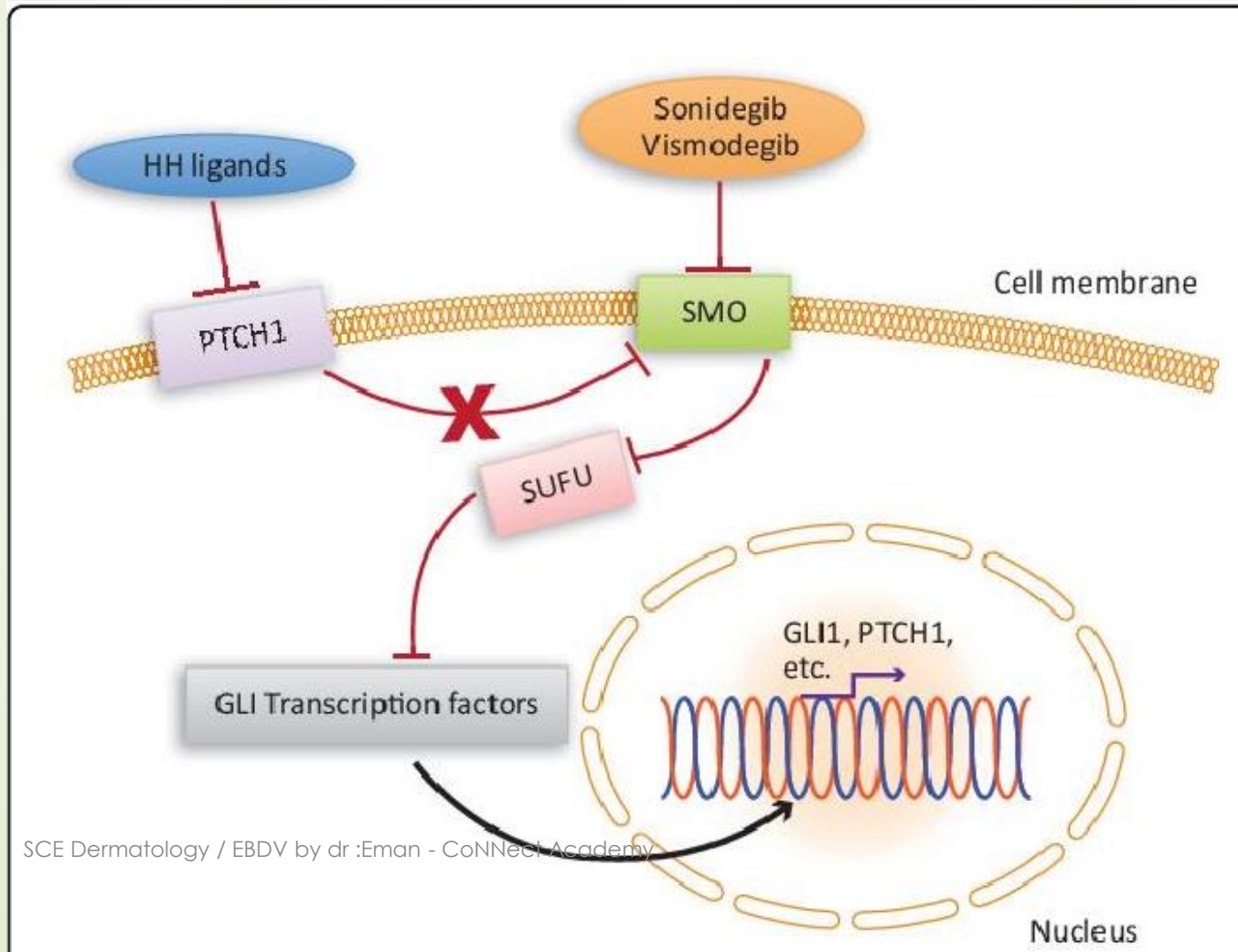
†More often trichoblastomas.

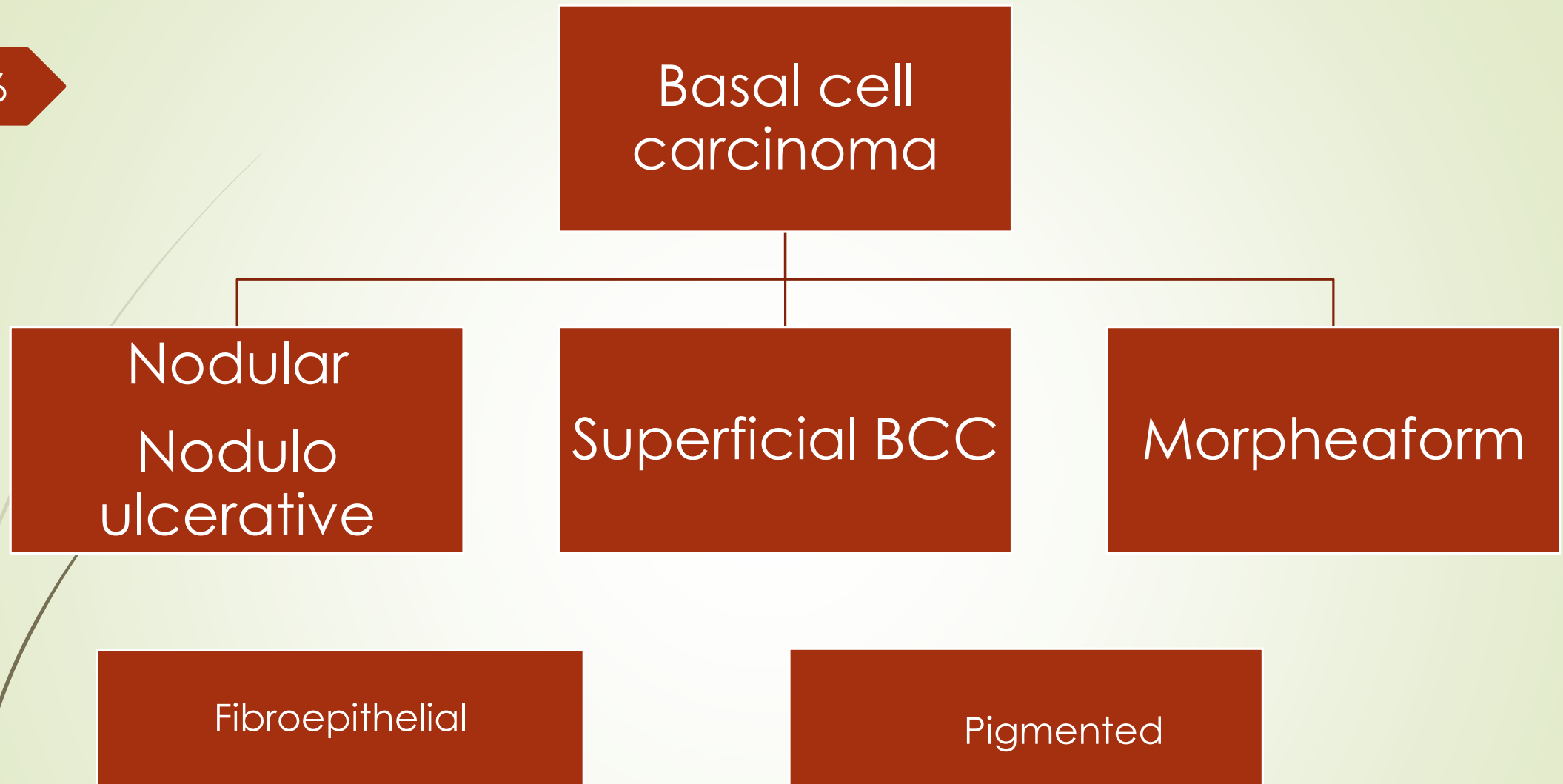
Basal cell carcinoma

- Most common skin cancer in human
Men 2; 1 elderly



HEDGEHOG signaling pathway





Nodular , noduloulcerative

- Most common 50%
- Head , neck , face
- Papule , nodule >>>> arborizing telangiectasia >>>>
 Ulcerate >>>>> rodent ulcer >>>>>
 elevated rolled border

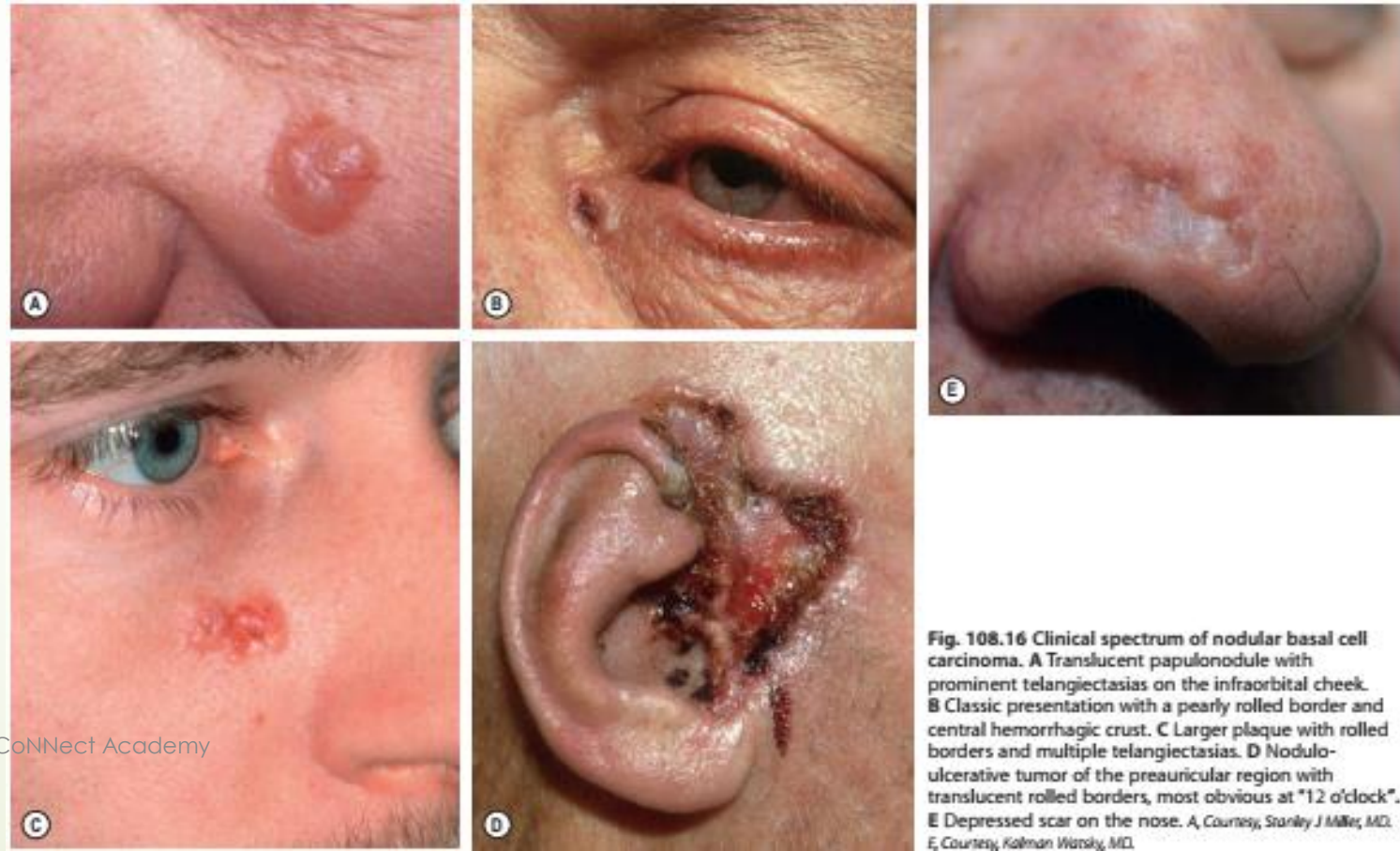
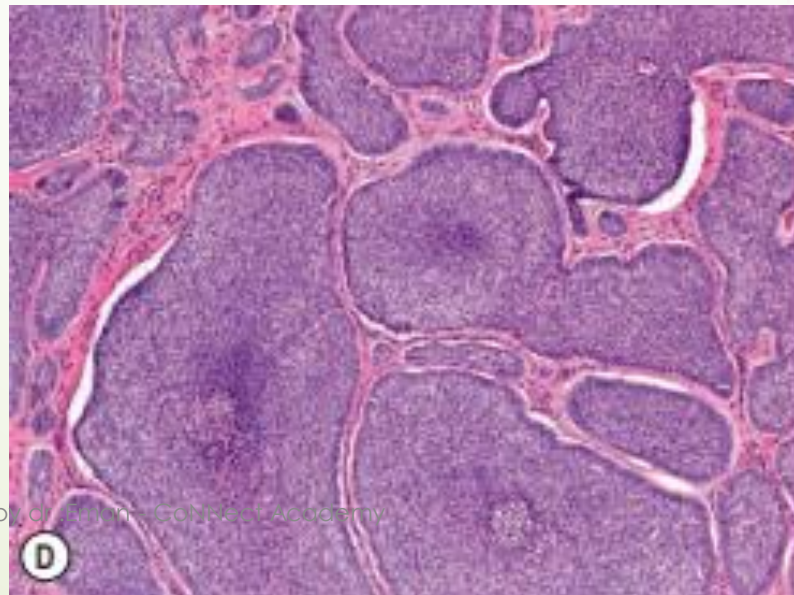
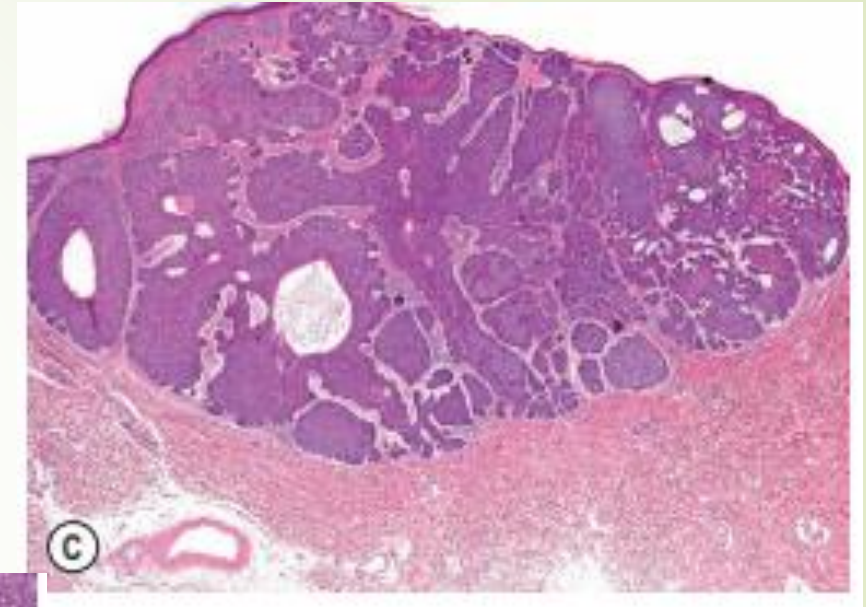


Fig. 108.16 Clinical spectrum of nodular basal cell carcinoma. A Translucent papulonodule with prominent telangiectasias on the infraorbital cheek. B Classic presentation with a pearly rolled border and central hemorrhagic crust. C Larger plaque with rolled borders and multiple telangiectasias. D Noduloulcerative tumor of the preauricular region with translucent rolled borders, most obvious at "12 o'clock". E Depressed scar on the nose. A, Courtesy, Stanley J Miller, MD. E, Courtesy, Kolman Warsky, MD.

Histeopathology

- Mass of basaloid cells surrounded by fibromyxoid stroma
- Peripheral palisading
- Retraction of stroma >>>> cleft
- Micronodular bcc : much smaller cell i
- Basaloid squamous carcinoma

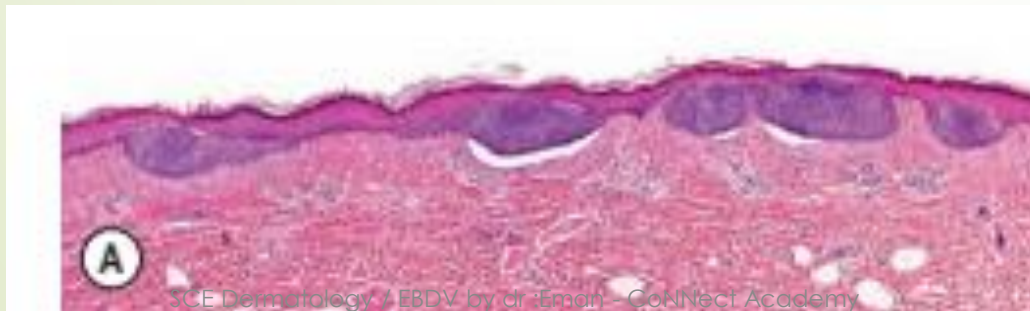


Superficial bcc

- Young age
- Trunk
- Scaly , erythematous well defined patch or thin plaque

With raised border (thread like) whip cord

- H.P :
 - Small , superficial located



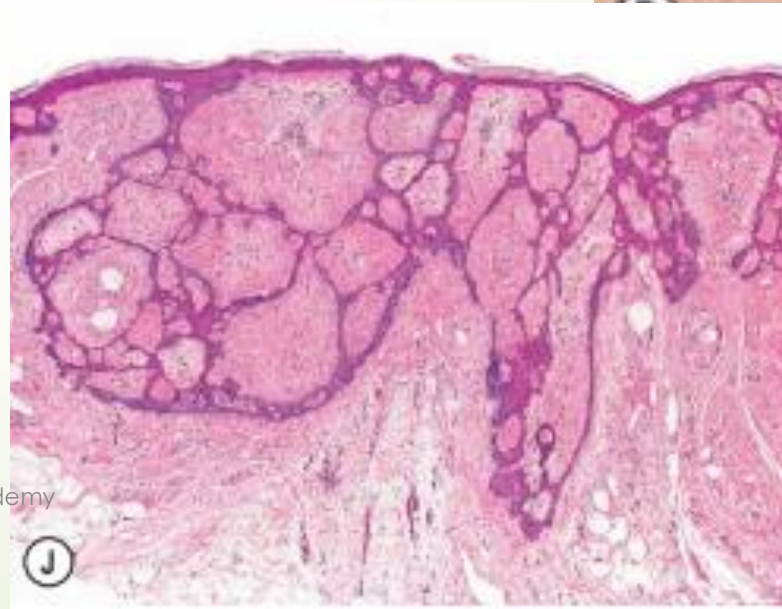
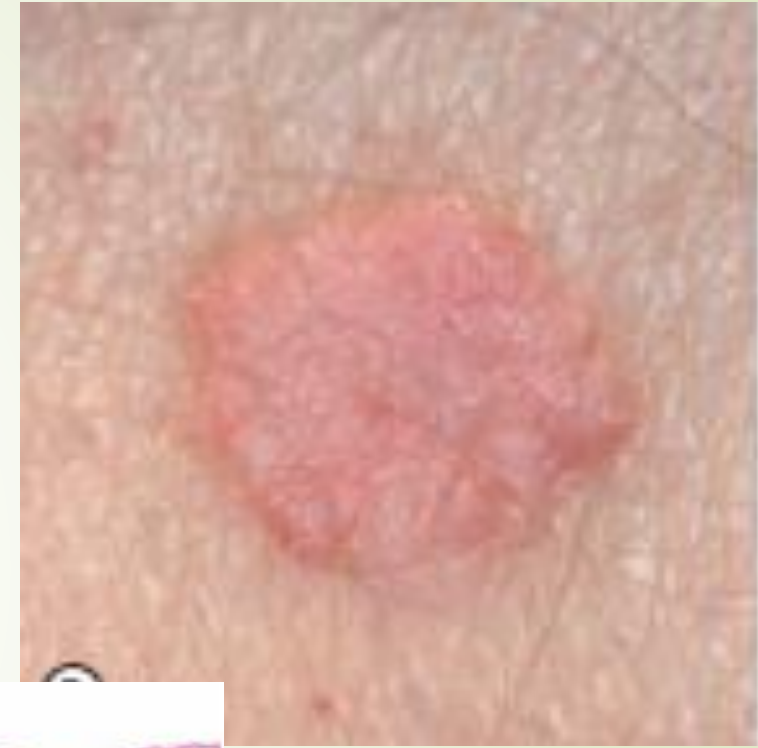
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Fig. 108.18 Superficial basal cell carcinomas. A Numerous erythematous patches and thin plaques on the back of a man with a history of arsenic exposure decades previously. B A solitary large, thin, dark pink plaque. There are scattered areas of fine scaling and small foci of brown pigment within the rolled border. As a rule, these lesions are neither pruritic nor tender.

Fibroepithelial

- Trunk
 - Lumbosacral
 - Sessile ,pedunculated
 - Soft
 - Skin tag
 - DD :
 - Skin tag
 - Intradermal melanocytic nevus
- Anastomosing strand , cord , fenestrated



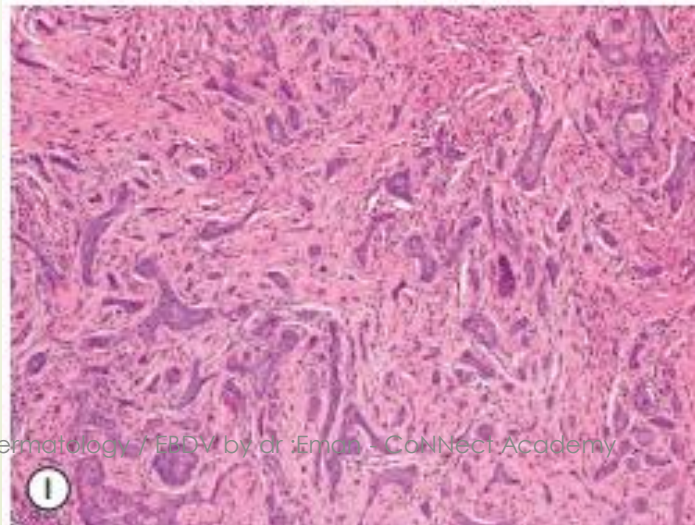
Pigmented bcc

- Dark skin
- DD : melanoma
- Hp : melanocyte , melanin , melanophages



Morpheaform

- Less common
- Indurated ivory white
- Ill defined border
- Smooth surface
- aggressive course
- DD :morphea , scar
- H.p :dens fibrosis , no palisading



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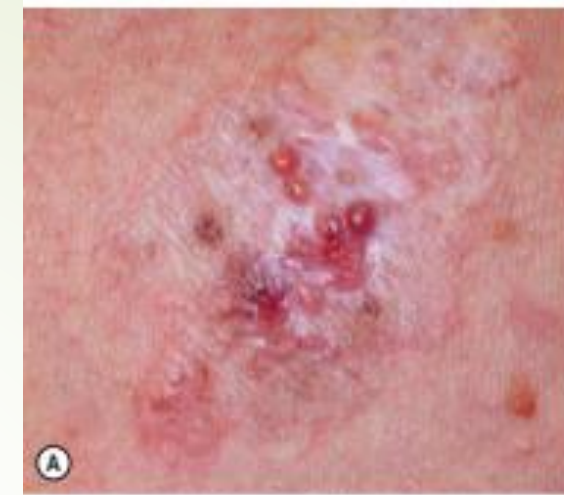


Fig. 108.19
Morpheaform basal cell carcinomas. **A** Recurrent tumor two years after microscopically controlled surgery; note the scar-like appearance with superimposed glassy pink and brown papules. **B** Oval hypopigmented firm plaque that resembles a scar (e.g. post electrodesiccation and curettage). While there is a light pink color between 6 and 9 o'clock, no translucency or rolled border is present. **C** A classic example with indistinct borders and scar-like appearance.

BCC syndromes

➔ Gorline(BCNS)

- ➔ AD ,
- ➔ Mutation in ptch 1
- ➔ Multiple early BCC



Fig. 108.1 Basal cell nevus syndrome.
A Numerous small and several large nodular BCCs on the face.
B Multiple palmar pits.
B, Courtesy, Jeffrey P Callen, MD.

DIAGNOSTIC CRITERIA FOR BASAL CELL NEVUS SYNDROME (BCNS)

Major criteria

1. More than two BCCs or one BCC before the age of 20 years
2. Odontogenic keratocysts of the jaw (proven by histology)
3. Three or more palmar or plantar pits
4. Bilamellar calcification of the falx cerebri
5. Bifid, fused or markedly splayed ribs
6. First-degree relative with BCNS

Minor criteria

1. Macrocephaly (determined after adjustment for height)
2. Congenital malformations: cleft lip or palate; frontal bossing; "coarse face"; moderate or severe hypertelorism
3. Other skeletal abnormalities: Sprengel deformity^{*}; marked pectus deformity; marked syndactyly of the digits
4. Radiographic abnormalities: bridging of the sella turcica; vertebral anomalies such as hemivertebrae and fusion or elongation of the vertebral bodies; modeling defects of the hands and feet; flame-shaped lucencies of the hands or feet
5. Bilateral ovarian fibromas
6. Medulloblastoma

➤ 2- bazex dupre- christol syndrome :

- Mutiple facial bcc
- Follicular Atrophoderma
- Hypotrichosis , hypohidrosis



DD bazex syndrome acrokeratosis para neoplastica



- 3- **ROMBO SYNDROM :**
 - Atrophoderma vermiculatum
 - Hypotrichosis , blepharitis
 - telangectasia
 - Trichepitheima
 - BCCS



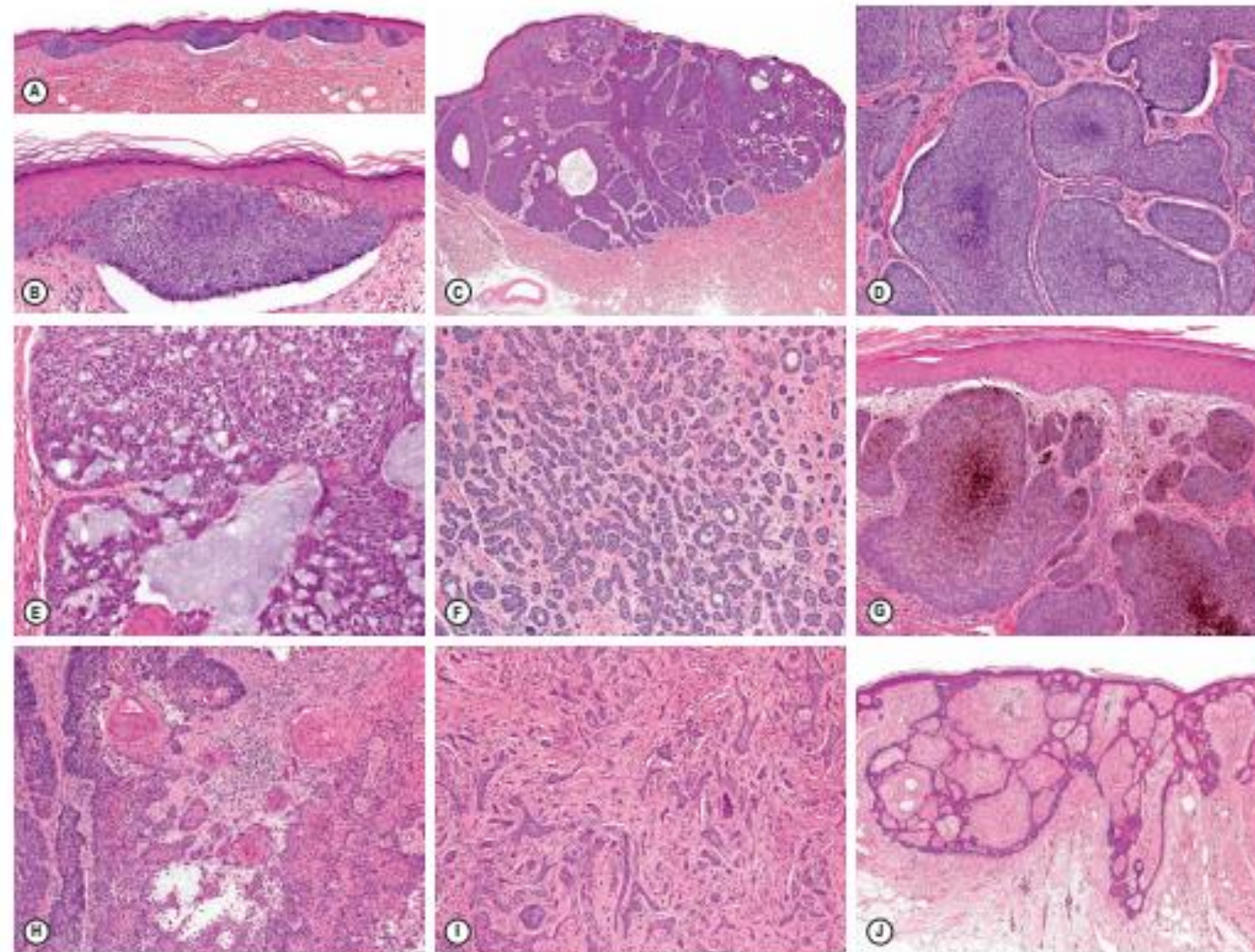
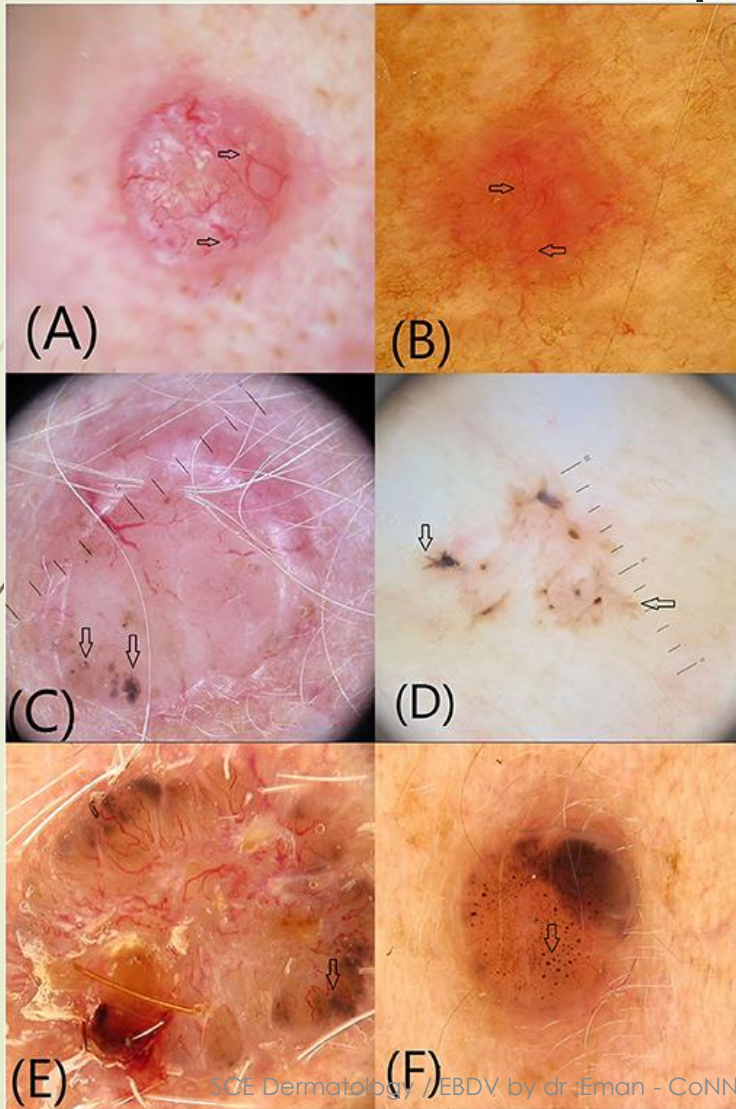
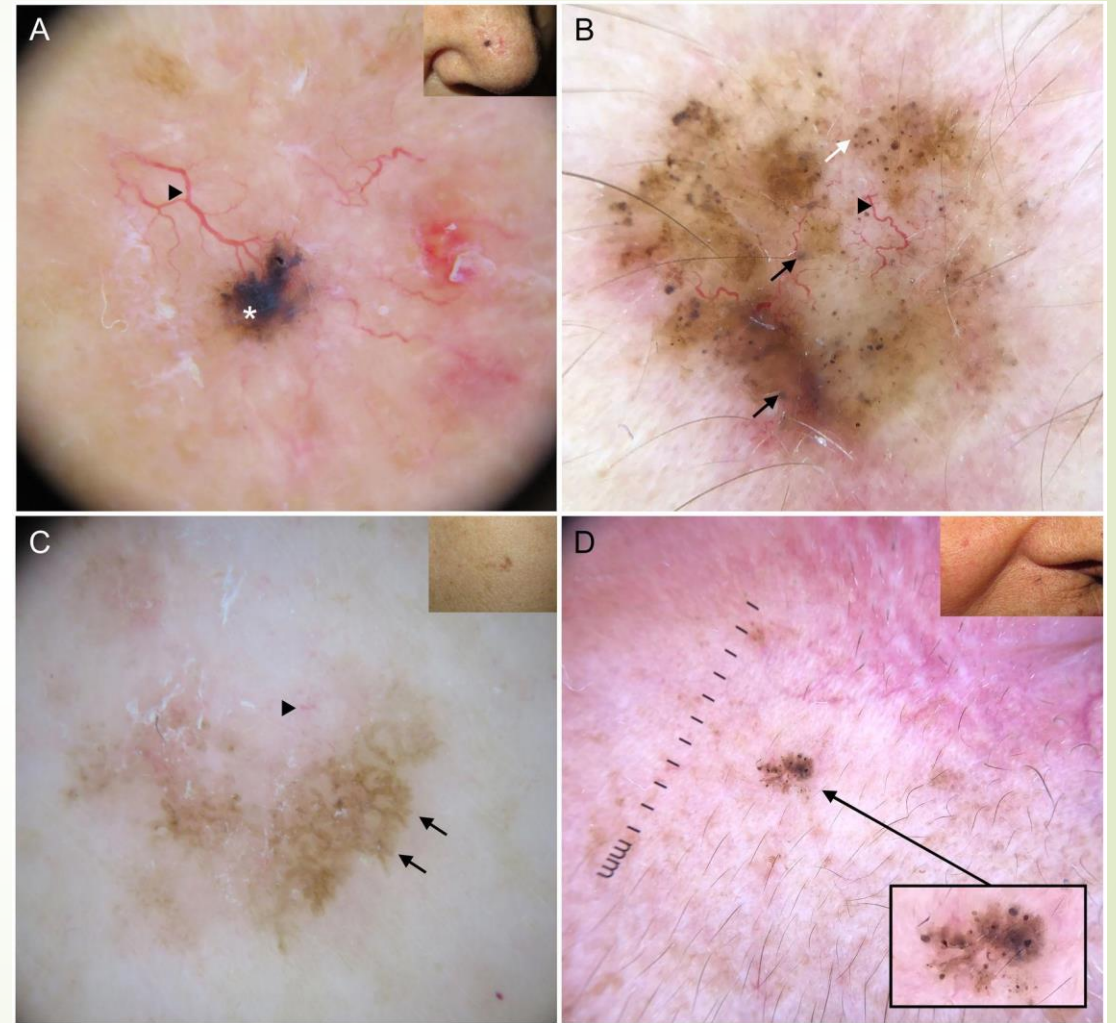
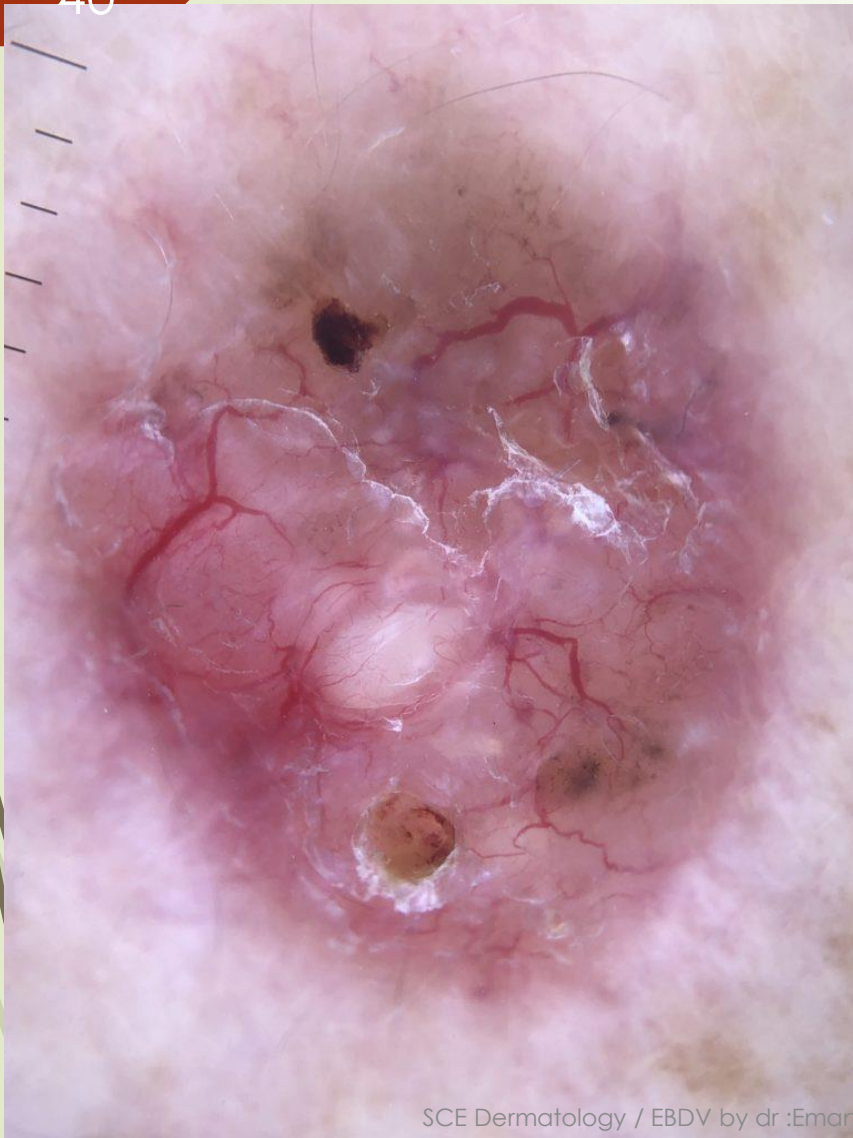


Fig. 108.23 Basal cell carcinoma (BCC): range of histopathologic findings. A Superficial type with a few basaloid buds and stromal retraction spaces. B A single basaloid aggregation with all the morphologic findings (e.g. peripheral palisading of nuclei, retraction between the tumor and surrounding stroma) characteristic of a superficial BCC. C Nodular BCC with large, partly confluent aggregations of basaloid cells as well as a few cystic spaces. D Nodular aggregations of basaloid cells with peripheral palisading and retraction spaces commonly seen in conventional nodular BCCs. E Nodular aggregations of basaloid cells with mucin deposition within the epithelial component, referred to as mucinous BCC. F Micronodular BCC characterized by numerous, densely packed, small nodular aggregations of basaloid cells. G Pigmented BCC with nodular aggregations of basaloid cells that contain brown melanin pigment. H Basosquamous BCC in which both basaloid cells with peripheral palisading and eosinophilic cells with slightly pleomorphic nuclei and foci of keratinization ("horn pearls") are present. I Morpheaform BCC with numerous basaloid strands and cords with an irregular jagged outline embedded within a fibrous stroma. J Fibroepithelial BCC (fibroepithelioma of Pinkus) with a fenestrated pattern of anastomosing epithelial cords. Courtesy, Lorenzo Cerroni, MD.

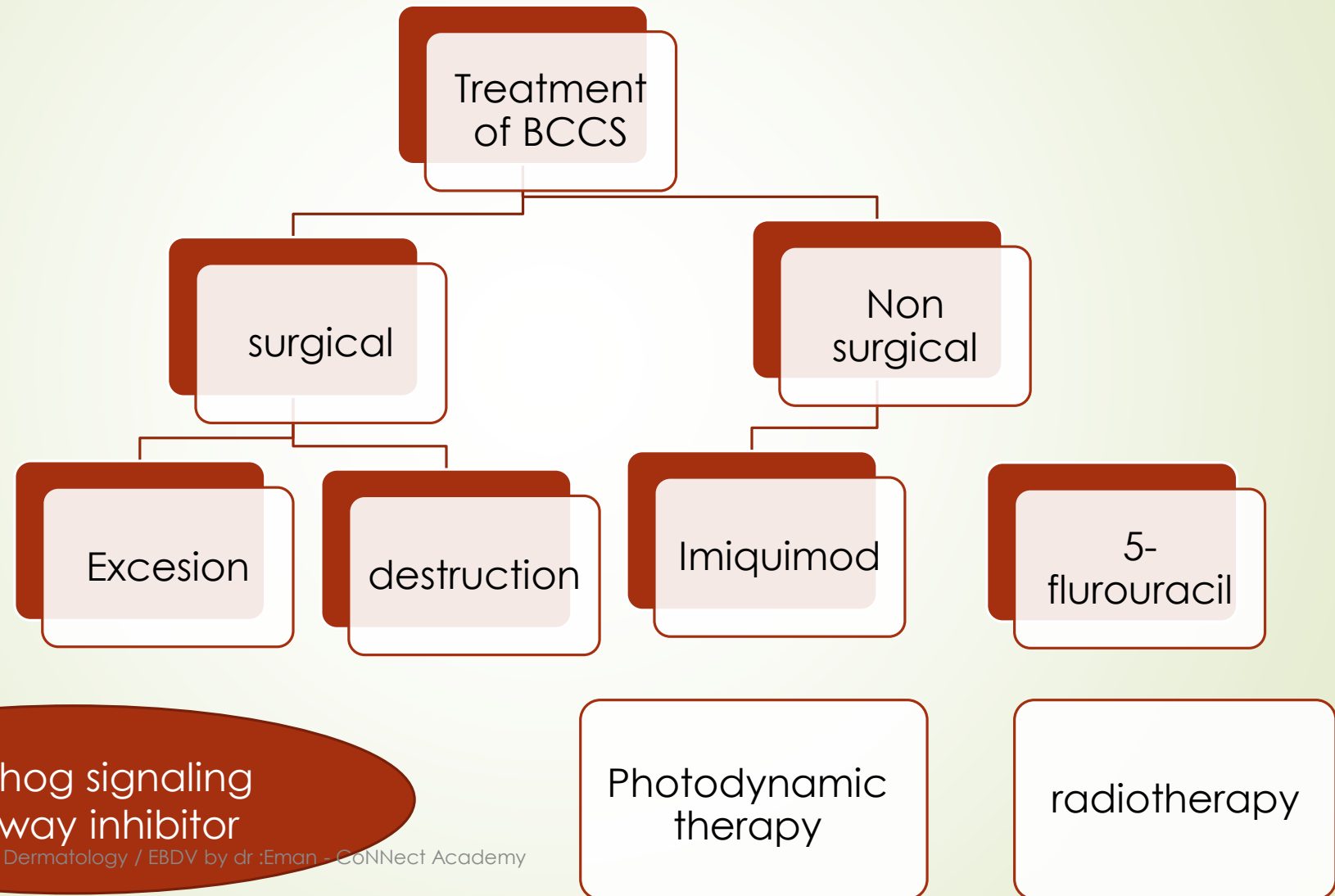
Dermoscopy of BCC



1. maple leaf
2. Spoke wheel
3. Arborizing telangiectasia
4. ulceration
5. Blue gray nest
6. Mutipl eblue gray globules



HISTOLOGIC SIMULANTS OF BASAL CELL CARCINOMA (BCC)	
	Major distinguishing features
Adenoid cystic carcinoma	Cribriform pattern; EMA-positive
Ameloblastoma	Location in mouth
Basaloid follicular hamartoma	Multiple small papules; positive family history
Cloacogenic carcinoma	Anal location
Dermatofibroma	Basaloid induction in overlying epidermis
Eccrine carcinoma	Small sweat ducts
Folliculocentric basaloid proliferation	Focal follicular budding
Merkel cell carcinoma	Punctate keratin (CK20-positive) in cytoplasm; positivity for Merkel cell polyomavirus
Metastatic breast carcinoma	History; single filing; expression of CK7, mammoglobin, gross cystic disease fluid protein-15
Microcystic adnexal carcinoma	Small keratin cysts, absent stromal retraction, ductal differentiation
Mucinous carcinoma	Pools of mucin with floating basaloid islands
Nevus sebaceus	Basaloid induction; trichoblastomas and BCCs may arise within a nevus sebaceus
Sebaceous carcinoma	Adipophilin-positive sebocytes
Trichoblastoma	Clefts within the stroma; hair germs and papillae; PHLDA1-positive
Trichoepithelioma and desmoplastic trichoepithelioma	Horn cysts, absent stromal retraction, papillary mesenchymal bodies; PHLDA1-positivity; peritumoral CD34 positivity; positive family history when multiple



Hedghog signaling
pathway inhibitor

Surgical excision

Standard surgical excision

1st line most effective treatment
4mm safety margins
2cm in morphea type

MOHS micrographic surgery

Maximal preservation of normal tissue

Destruction

Curettage
Curettage + cautery
Cryo
Laser: CO₂, PDL

NON SURGICAL

topical pharmacotherapy of skin cancer

1- Imiquimod

2,5%, 5%

FDA :AK , SUPERFICIAL
BCC, genital wart

IMMUNE MODIFIER
TLR7-8 agonist

Category c

2- 5- fluolouracil

.5% , 1%

FDA: AK , super-BCC

**Inhibition thymidylate
synthase >>>**
Reduse dna proliferation

Category x

**3- diclofenac
4- Ingenol
mebutate
5- retinoids**

- Hedgehog signaling pathway inhibitor

vismodegib 150mg

sonidegib 200 mg

inhibit smoothened pathway

SIDE EFFECTS OF VISMODEGIB AND SONIDEGIB	
Common	
• Dysguesia* (>40% of patients) • Muscle spasms • Alopecia	• Nausea, reduced appetite, and weight loss • Diarrhea • Fatigue
Less common	
• SCCs, including phenotypic switch of BCC to SCC • Cough • Congestive heart failure • Pneumonia • Thromboembolism	• Keratoacanthomas • Amenorrhea • Hyponatremia • Arthralgia • Pruritus

COMMON TREATMENT OPTIONS FOR ACTINIC KERATOSES		
Localized/lesion-targeted treatments*		
Lesion-targeted treatment		Comments
Liquid nitrogen cryosurgery		• No cutting or anesthesia necessary • 10- to 14-day healing period • Risk of hypopigmentation
Curettage (> curettage + electrodesiccation)		• Requires local anesthesia • Risk of hypopigmentation and scarring (less so if curettage alone)
Shave excision		• Requires local anesthesia • Risk of hypopigmentation and scarring
Topical field treatments***†		
Although these are the recommended FDA dosing schedules, if the patient develops significant burning with application or significant erosive erythema, the treatment should be held or discontinued		
Topical agent	Dosing	Comments
5-Fluorouracil	• 0.5% cream = apply in the evening x 4 weeks • 1% or 5% cream = apply twice a day for 2-4 weeks • 2% and 5% solutions = apply twice a day for 2-4 weeks • When treating extensor extremities, pretreatment with a topical retinoid for 1-2 weeks may be required or occluded "chemowraps" may be used	• Warn of photosensitivity, which can be severe • Optimal results occur if treatment continues until there is significant inflammation or superficial erosions • Healing usually occurs within 2 weeks of stopping treatment • Solution is best for hair-bearing areas, such as the scalp
Imiquimod	• 2.5% or 3.75% cream = apply at bedtime for 2 weeks; take a 2-week break, and then repeat 2-week application cycle • 5% cream = apply twice a week for 16 consecutive weeks • Recommended treatment area is 25 cm²	• May cause systemic flu-like symptoms • Not recommended if patient has an underlying autoimmune condition • May cause hypopigmentation at the treated site
Diclofenac	• 3% gel = apply twice a day for 90 days • A maximum amount of 8 g daily should not be exceeded	• Do not use if patient is allergic to nonsteroidal anti-inflammatory drugs (NSAIDs) • Although it may have a less severe cutaneous reaction, compliance is an issue given a longer application period
Ingenol mebutate	• 0.015% gel = apply at bedtime for 3 consecutive nights (face and scalp) • 0.05% gel = apply at bedtime for 2 consecutive nights (trunk and extremities)	• Rapid onset of action; typically within 24 hours of application, patient will experience erythema and burning • Healing occurs within 10-14 days • Supposedly does not cause hypopigmentation
Procedural field treatments***†		
Procedure		Comments
Photodynamic therapy (PDT; e.g. 5-ALA + blue light)		• Can be painful • Requires 48 hours of no outdoor exposure post-treatment • Relatively quick recovery over 1-2 weeks
Chemical peels (e.g. trichloroacetic acid)		• May require local anesthesia • May cause significant irritation and temporary discoloration • Healing typically over 7 days
Ablative laser techniques (e.g. ablative fractional lasers, erbium:YAG laser)		• Requires local anesthesia • Depending on the technique, recovery period varies • Hypopigmentation a potential complication

THANK
YOU!